Ryan Zhang

New York, NY | rz2716@columbia.edu | (917) 929-8106 | rhyzhang.github.io | rhyzhang | rhyzhang

Education

Columbia University — *New York, NY*

Bachelor of Arts in Computer Science | John Jay Scholar (Top 10%) | GPA 3.926 — *Sept 2024 – May 2028*

- **Coursework:** Honors Stats ML (In Progress), Advanced Programming (Systems), Prob & Stats, Data Structures.

Experience

Junior Researcher — *New York, NY*

Machine Learning for Good Lab, NYU Courant — *July 2023 – present*

- Researching human decision-making bias by developing a novel Bayesian model that simultaneously predicts outcomes and quantifies uncertainty.
- Accelerated model inference speed by 60x by linearizing key functions, improving experimental iteration speed by 10x.
- Designed and maintain the complete research ML pipeline of model implementation, data collection, training, and visualizations.
- Executing 3+ experiments per week leveraging High Performance Computing workflows in preparation for a first author publication.

Co-Director — *New York, NY*

Columbia AI Alignment Club, Columbia University — *Jan 2025 – present*

- Secured $75,000 in competitive grants to fund technical research tracks, enabling 150+ students to access high level AI safety resources and mentorship.
- Facilitated technical analysis of frontier LLM capabilities, leading advanced reading groups on mechanistic interpretability, RLHF, and scalable oversight.
- Selected for exclusive retreats and symposiums (OASIS @ Constellation, UChicago), directly engaging with researchers from Anthropic and top labs to understand emerging risks in foundation models.

Data Scientist — *New York, NY*

Rebecca & Christopher Cleaners — *Jan 2024 – present*

- Architected an end-to-end automated ETL pipeline, migrating the business from intuition based operations to data driven decision making by consolidating fragmented data sources.
- Reverse engineered undocumented legacy POS schemas (SQL Server, MDB) and developed custom scripts to extract, sanitize, and load sales data into a centralized data lake.
- Developed a Streamlit analytics dashboard to visualize KPIs and revenue patterns, identifying opportunities during the holiday season to optimize operations scheduling.

Projects

LionTalk — *Dec 2025 – present*

- Technologies: React (Next.js), Python, Docker, Vercel, GitHub Actions, CI/CD, LLM Integration
- Collaborated on full-stack development of a centralized platform for displaying seminars around campus; Implementing a Next.js frontend and coordinating with the backend to define JSON data schemas to enable parallel development.
- Engineered an automated data pipeline by containerizing Python-based LLM scrapers with Docker and deploying them with GitHub Actions to ensure consistent and reliable data updates.

HTTP Web Server & Database Engine — *Nov 2025 – Dec 2025*

- Technologies: C, BSD Sockets, TCP/IP, Linux, Make, Valgrind
- Architected a concurrent HTTP 1.0 web server and a standalone database server from scratch in C, utilizing BSD sockets for robust client-server communication.
- Implemented a reverse-proxy architecture to handle static file serving and dynamic database queries, ensuring strict memory safety and zero leaks via Valgrind.

Technical Skills

Languages: Python, C, Java, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Streamlit, pandas, NumPy, scikit-learn, Matplotlib, Seaborn, pytest

Machine Learning/Data: MLflow, DVC, Jupyter Notebooks

DevOps & Tools: Git, Docker, GitHub Actions (CI/CD), Linux/WSL, HPC, Slurm, Conda, Make, Valgrind